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Modular test system

Abstract

A modular test system includes a modular controller subsystem coupled to a modular tested subsystem by a modular intermediary subsystem. The modular intermediary subsystem transmits communications between the modular controller subsystem and the modular tested subsystem during testing operations performed by the modular controller subsystem, and performs at least one testing function that is not available in the modular controller subsystem during the testing operations performed by the modular controller subsystem.

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Background/Summary

BACKGROUND

(1) The present disclosure relates generally to information handling systems, and more particularly to a modular test system for testing subsystems utilized in information handling systems.

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

(3) Information handling systems such as, for example, server devices, networking devices (e.g., switch devices), storage systems, desktop computing devices, laptop/notebook computing devices, tablet computing devices, mobile phones, and/or other computing devices known in the art, often include subsystems that are tested during their development and/or otherwise prior to being provided in those computing devices. For example, storage devices such as Solid State Drive

(SSD) storage devices including NAND storage subsystems provide just one example of subsystems that may be included in such computing devices and subject of the testing described below. The testing of subsystems in computing devices such as Integrated Circuit (IC) components like the NAND storage subsystems/storage media discussed above may include the testing of a variety of different storage media usages and functions. In specific examples, high-speed data transmission testing that utilizes relatively complex command sets with relatively intricate timing requirements may be performed on the storage media, power consumption testing may be performed for any of a variety of operations associated with the storage media, temperature-dependent behavior testing may be performed to measure the behavior of the storage media at different temperatures, and data retention testing may be performed to determine data retention capabilities under different usage and temperature conditions. As will be appreciated by one of skill in the art, significant variation in the result of such testing may exist between different storage media (e.g., across different NAND technology generations, manufacturers, etc.).

(4) One of skill in the art will appreciate how the different testing described above requires physically different measurement systems and pathways. For example, for the high-speed data transmission testing described above, it is desirable to provide high-speed data connections on a controller that connect to the storage media in order to test maximum speed capabilities, efficiently execute test instructions, and/or accomplish other high-speed testing objectives that would be apparent to one of skill in the art. However, in other examples, it may be desirable to provide debug ports coupled to a communication bus on the controller that connects the controller to the storage media in order to allow for the analysis of commands and data exchanged with the storage media, and one of skill in the art in possession of the present disclosure will appreciate how such debug ports would interfere with the high-speed data transmission testing described above, thus requiring different controllers with different configurations (e.g., different controller paths, different signal turning parameters, etc.) in order to perform the high-speed data transmission testing and the debug testing discussed above.

(5) Furthermore, some conventional storage media testing techniques utilize actual SSD storage devices or SSD-storage-device-like configurations that may include the storage media under test (e.g., providing NAND storage systems in an existing storage device/storage device chassis or on a chassis having an existing storage device configuration), and that may be connected to an SSD controller IC or Field Programmable Gate Array (FPGA)-type controller that is configured to test different characteristics of that storage media. Further still, in some situations, “test couplings” (e.g., test probes) may be provided in the storage device or storage-device-like configuration to capture particular data (e.g., voltage data, current data, etc.), perform data analysis (e.g., debug data analysis), and/or other testing operations that would be apparent to one of skill in the art. As such, based on the type of testing being conducted, conventional storage media testing systems may provide either or both of the controller and storage media/storage device with a particular configuration or design that includes the features needed to perform that type of testing.

(6) As will be appreciated by one of skill in the art, the conventional storage media test systems discussed above, and the associated time to perform such testing, is relatively expensive, as each physical configuration or design (e.g., to perform high speed data transmission testing, debug operations, power measurement, etc.) requires particular resources and resource configurations to accomplish the associated testing objectives. Furthermore, controlling circuitry and storage media generations may evolve out of phase with each other, and the storage-media-specific configuration/design of conventional storage media testing systems requires newly configured/designed storage media testing systems when new storage media is developed.

(7) Accordingly, it would be desirable to provide a test system that addresses the issues discussed above.

SUMMARY

(8) According to one embodiment, a modular intermediary subsystem includes a modular

intermediary board; a first modular intermediary connector that is included on the modular intermediary board and that is configured to connect to a modular controller subsystem; a second modular intermediary connector that is included on the modular intermediary board and that is configured to connect to a modular tested subsystem; a first modular intermediary board subsystem that is included on the modular intermediary board and that is configured to: transmit communications between the modular controller subsystem and the modular tested subsystem during testing operations performed by the modular controller subsystem; and a second modular intermediary board subsystem that is included on the modular intermediary board and that is configured to: perform at least one testing function that is not available in the modular controller subsystem during the testing operations performed by the modular controller subsystem.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic view illustrating an embodiment of an Information Handling System (IHS).
- (2) FIG. 2 is a schematic view illustrating an embodiment of a modular test system provided according to the teachings of the present disclosure and coupled to one or more testing devices.
- (3) FIG. 3 is a schematic view illustrating an embodiment of a modular controller subsystem that may be included in the modular testing system of FIG. 2.
- (4) FIG. 4 is a schematic view illustrating an embodiment of a modular tested subsystem provided by a modular media subsystem that may be included in the modular test system of FIG. 2.
- (5) FIG. 5 is a schematic view illustrating an embodiment of a modular intermediary subsystem provided by a modular power measuring subsystem that may be included in the modular test system of FIG. 2.
- (6) FIG. 6 is a schematic view illustrating an embodiment of a modular intermediary subsystem provided by a modular debug data analyzing subsystem that may be included in the modular test system of FIG. 2.
- (7) FIG. 7 is a schematic view illustrating an embodiment of a modular intermediary subsystem provided by a modular expansion subsystem that may be included in the modular test system of FIG. 2.
- (8) FIG. 8 is a schematic view illustrating an embodiment of a modular temperature measurement subsystem that may be included in the modular test system of FIG. 2.
- (9) FIG. 9 is a schematic view illustrating an embodiment of a modular intermediary subsystem provided by a modular PCIe connect subsystem that may be included in the modular test system of FIG. 2.
- (10) FIG. 10 is a flow chart illustrating an embodiment of a method for modular testing of a tested subsystem.
- (11) FIG. 11 is a schematic view illustrating an embodiment of the modular controller subsystem of FIG. 3 connected to the modular media subsystem of FIG. 4 to provide the modular test system of FIG. 2.
- (12) FIG. 12 is a schematic view illustrating an embodiment of the modular controller subsystem of FIG. 3 connected to the modular media subsystem of FIG. 4 by the modular power measuring subsystem of FIG. 5 to provide the modular test system of FIG. 2 during the method of FIG. 10.
- (13) FIG. 13 is a schematic view illustrating an embodiment of the modular controller subsystem of FIG. 3 connected to the modular media subsystem of FIG. 4 by the modular debug data analyzing subsystem of FIG. 6 to provide the modular test system of FIG. 2 during the method of FIG. 10.
- (14) FIG. 14 is a schematic view illustrating an embodiment of the modular controller subsystem of FIG. 3 connected to a plurality of the modular media subsystem of FIG. 4 by the modular

expansion subsystem of FIG. 7 to provide the modular test system of FIG. 2 during the method of FIG. 10.

(15) FIG. 15 is a schematic view illustrating an embodiment of the modular controller subsystem of FIG. 3 connected to the modular media subsystem of FIG. 4 by the modular PCIe connect subsystem of FIG. 9 to provide the modular test system of FIG. 2 during the method of FIG. 10.

(16) FIG. 16 is a schematic view illustrating an embodiment of the modular temperature measurement subsystem of FIG. 7 connected to a plurality of the modular media subsystem of FIG. 4 to provide the modular test system of FIG. 2.

DETAILED DESCRIPTION

(17) For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., personal digital assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

(18) In one embodiment, IHS 100, FIG. 1, includes a processor 102, which is connected to a bus 104. Bus 104 serves as a connection between processor 102 and other components of IHS 100. An input device 106 is coupled to processor 102 to provide input to processor 102. Examples of input devices may include keyboards, touchscreens, pointing devices such as mice, trackballs, and trackpads, and/or a variety of other input devices known in the art. Programs and data are stored on a mass storage device 108, which is coupled to processor 102. Examples of mass storage devices may include hard discs, optical disks, magneto-optical discs, solid-state storage devices, and/or a variety of other mass storage devices known in the art. IHS 100 further includes a display 110, which is coupled to processor 102 by a video controller 112. A system memory 114 is coupled to processor 102 to provide the processor with fast storage to facilitate execution of computer programs by processor 102. Examples of system memory may include random access memory (RAM) devices such as dynamic RAM (DRAM), synchronous DRAM (SDRAM), solid state memory devices, and/or a variety of other memory devices known in the art. In an embodiment, a chassis 116 houses some or all of the components of IHS 100. It should be understood that other buses and intermediate circuits can be deployed between the components described above and processor 102 to facilitate interconnection between the components and the processor 102.

(19) Referring now to FIG. 2, an embodiment of a testing system 200 is illustrated. In the illustrated embodiment, the testing system 200 includes a modular test system 202 that is provided according to the teachings of the present disclosure, and that is discussed in further detail below. The modular test system 202 is coupled to one or more testing device(s) 204. In an embodiment, any of the test device(s) 204 may be provided by the IHS 100 discussed above with reference to FIG. 1, and/or may include some or all of the components of the IHS 100, and in the specific examples provided herein are described as desktop computing devices, laptop/notebook computing devices, power management data analysis computing devices, and debug data analysis computing devices. However, while illustrated and discussed as being provided by particular computing

devices, one of skill in the art in possession of the present disclosure will recognize that test devices provided in the testing system **200** may include any devices or systems (e.g., a temperature-controlled environment system that may perform the storage media “baking” operations described below) that may be configured to perform the testing discussed below. However, while a specific testing system **200** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that the modular test system of the present disclosure may be used in variety of testing systems while remaining within the scope of the present disclosure as well.

(20) Referring now to FIG. 3, an embodiment of a modular controller subsystem **300** is illustrated that may be included in the modular test system **202** discussed above with reference to FIG. 2. In the illustrated embodiment, the modular controller subsystem **300** includes a chassis **302** that supports the components of the modular controller subsystem **300**, only some of which are illustrated and discussed below. For example, the chassis **302** may be provided by a circuit board **302**, although other chassis are envisioned as falling within the scope of the present disclosure as well. A controller device **304** is mounted to the chassis **302** and, in some examples, may include a processing system (not illustrated, but which may include the processor **102** discussed above with reference to FIG. 1) and a memory system (not illustrated, but which may include the memory **114** discussed above with reference to FIG. 1) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide a controller engine that is configured to perform the functionality of the controller engines and/or controller devices discussed below.

(21) A plurality of connectors **306a** and **306b** are included on the chassis **302**, and one of skill in the art in possession of the present disclosure will appreciate how the connectors **306a** and **306b** may be coupled to the controller device **304** via the chassis **302**. For example, in embodiments in which the chassis **302** is a circuit board, that circuit board may include traces, wiring, couplings, and/or other circuit board subsystems that couple the connectors **306a** and **306b** to the controller device **304**. However, while a specific modular controller subsystem **300** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that modular controller subsystems (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the modular controller subsystem) may include a variety of components and/or component configurations for providing the functionality discussed below while remaining within the scope of the present disclosure as well.

(22) Referring now to FIG. 4, an embodiment of a modular tested subsystem provided by a modular media subsystem **400** is illustrated that may be included in the modular test system **202** discussed above with reference to FIG. 2. In the illustrated embodiment, the modular media subsystem **400** includes a chassis **402** that supports the components of the modular media subsystem **400**, only some of which are illustrated and discussed below. For example, the chassis **402** may be provided by a circuit board, although other chassis are envisioned as falling within the scope of the present disclosure as well. A plurality of media devices **404** are mounted to the chassis **402** and, in some examples, may be provided by storage media such as NAND storage subsystems. However, while a particular modular tested subsystem including particular tested devices (e.g., storage media devices provided by NAND storage subsystems in the illustrated examples) is provided, one of skill in the art in possession of the present disclosure will recognize how a variety of modular tested subsystems may provide different tested devices for testing according to the teachings of the present disclosure while remaining within its scope as well.

(23) A connector **406** is included on the chassis **402**, and one of skill in the art in possession of the present disclosure will appreciate how the connector **406** may be coupled to the media devices **404** via the chassis **402**. For example, in embodiments in which the chassis **402** is a circuit board, that circuit board may include traces, wiring, couplings, and/or other circuit board subsystems that couple the connector **406** to the media devices **404**. However, while a specific modular tested subsystem provided by a modular media subsystem **400** has been illustrated and described, one of

skill in the art in possession of the present disclosure will recognize that modular tested subsystems (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the modular media subsystem **400**) may include a variety of components and/or component configurations for providing the functionality discussed below while remaining within the scope of the present disclosure as well.

(24) Referring now to FIG. 5, an embodiment of a modular intermediary subsystem provided by a modular power measuring subsystem **500** is illustrated that may be included in the modular test system **202** discussed above with reference to FIG. 2. In the illustrated embodiment, the modular power measuring subsystem **500** includes a chassis **502** that supports the components of the modular power measuring subsystem **500**, only some of which are illustrated and discussed below. For example, the chassis **502** may be provided by a circuit board, although other chassis are envisioned as falling within the scope of the present disclosure as well. A plurality of connectors **504a**, **504b**, and **504c** are included on the chassis **502**, and one of skill in the art in possession of the present disclosure will appreciate how the connectors **504a**, **504b**, and **504c** may be coupled to each other via the chassis **502**. For example, in embodiments in which the chassis **502** is a circuit board, that circuit board may include traces, wiring, couplings, and/or other circuit board subsystems that couple the connectors **504a**, **504b**, and **504c** to each other.

(25) In the specific examples described below, the chassis **502**/circuit board includes a first modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit communications between the connectors **504a** and **504b** during testing operations. Furthermore, in some embodiments, the chassis **502**/circuit board may include a second modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit data received at the connector **504b** to the connector **504c**. Furthermore, in some embodiments, the chassis **502**/circuit board may include a second modular intermediary board subsystem that may be configured to provide processing functionality that performs a power measurement function on data received via the connector **504b**, and generates power measurement data that it transmits via the connector **504a**.

(26) As such, while not explicitly illustrated in FIG. 5, in some embodiments the chassis **502** may support a processing system (not illustrated, but which may include the processor **102** discussed above with reference to FIG. 1) and a memory system (not illustrated, but which may include the memory **114** discussed above with reference to FIG. 1) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide a power measuring engine that is configured to perform the functionality of the power measuring engines and/or modular power measuring subsystems discussed below, and that one of skill in the art in possession of the present disclosure will appreciate may be coupled to any or all of the connectors **504a-504c**. However, while a specific modular power measuring subsystem **500** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that modular intermediary subsystems (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the modular power measuring subsystem **500**) may include a variety of components and/or component configurations for providing the functionality discussed below while remaining within the scope of the present disclosure as well.

(27) Referring now to FIG. 6, an embodiment of a modular intermediary subsystem provided by a modular debug data analyzing subsystem **600** is illustrated that may be included in the modular test system **202** discussed above with reference to FIG. 2. In the illustrated embodiment, the modular debug data analyzing subsystem **600** includes a chassis **602** that supports the components of the modular debug data analyzing subsystem **600**, only some of which are illustrated and discussed below. For example, the chassis **602** may be provided by a circuit board, although other chassis are envisioned as falling within the scope of the present disclosure as well. A plurality of connectors

604a, **604b**, and **604c** are included on the chassis **602**, and one of skill in the art in possession of the present disclosure will appreciate how the connectors **604a**, **604b**, and **604c** may be coupled to each other via the chassis **602**. For example, in embodiments in which the chassis **602** is a circuit board, that circuit board may include traces, wiring, couplings, and/or other circuit board subsystems that couple the connectors **604a**, **604b**, and **604c** to each other.

(28) In the specific examples described below, the chassis **602**/circuit board includes a first modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit communications between the connectors **604a** and **604b** during testing operations. Furthermore, in some embodiments, the chassis **602**/circuit board may include a second modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit data received at the connector **604b** to the connector **604c**. Furthermore, in some embodiments, the chassis **602**/circuit board may include a second modular intermediary board subsystem that may be configured to provide processing functionality that performs a debug data analyzing function on data received via the connector **604b**, and generates debug data that it transmits via the connector **604a**.

(29) As such, while not explicitly illustrated in FIG. 6, in some embodiments the chassis **602** may support a processing system (not illustrated, but which may include the processor **102** discussed above with reference to FIG. 1) and a memory system (not illustrated, but which may include the memory **114** discussed above with reference to FIG. 1) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide a debug data analyzing engine that is configured to perform the functionality of the debug data analyzing engines and/or modular debug data analyzing subsystems discussed below, and that one of skill in the art in possession of the present disclosure will appreciate may be coupled to any or all of the connectors **604a-604c**. However, while a specific modular debug data analyzing subsystem **600** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that modular intermediary subsystems (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the modular debug data analyzing subsystem **600**) may include a variety of components and/or component configurations for providing the functionality discussed below while remaining within the scope of the present disclosure as well.

(30) Referring now to FIG. 7, an embodiment of a modular intermediary subsystem provided by a modular expansion subsystem **700** is illustrated that may be included in the modular test system **202** discussed above with reference to FIG. 2. In the illustrated embodiment, the modular expansion subsystem **700** includes a chassis **702** that supports the components of the modular expansion subsystem **700**, only some of which are illustrated and discussed below. For example, the chassis **702** may be provided by a circuit board, although other chassis are envisioned as falling within the scope of the present disclosure as well. A plurality of connectors **704a**, **704b**, and **704c** are included on the chassis **702**, and one of skill in the art in possession of the present disclosure will appreciate how the connectors **704a**, **704b**, and **704c** may be coupled to each other via the chassis **702**. For example, in embodiments in which the chassis **702** is a circuit board, that circuit board may include traces, wiring, couplings, and/or other circuit board subsystems that couple the connectors **704a**, **704b**, and **704c** to each other.

(31) In the specific examples described below, the chassis **702**/circuit board includes a first modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit communications between the connector **704a** and the connectors **704b** and **704c** during testing operations. Furthermore, in some embodiments, the chassis **702**/circuit board may include a second modular intermediary board subsystem that may be configured to provide processing functionality that performs expansion data routing functions on data transmitted between any of the connectors **704a**, **704b**, and/or **704c**.

(32) As such, while not explicitly illustrated in FIG. 7, in some embodiments the chassis **702** may support a processing system (not illustrated, but which may include the processor **102** discussed above with reference to FIG. 1) and a memory system (not illustrated, but which may include the memory **114** discussed above with reference to FIG. 1) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide an expansion data routing engine that is configured to perform the functionality of the expansion data routing engines and/or modular expansion subsystems discussed below, and that one of skill in the art in possession of the present disclosure will appreciate may be coupled to any or all of the connectors **704a-704c**. However, while a specific modular expansion subsystem **700** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that modular intermediary subsystems (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the modular expansion subsystem **700**) may include a variety of components and/or component configurations for providing the functionality discussed below while remaining within the scope of the present disclosure as well.

(33) Referring now to FIG. 8, an embodiment of a modular intermediary subsystem provided by a modular temperature measurement subsystem **800** is illustrated that may be included in the modular test system **202** discussed above with reference to FIG. 2. In the illustrated embodiment, the modular temperature measurement subsystem **800** includes a chassis **802** that supports the components of the modular temperature measurement subsystem **800**, only some of which are illustrated and discussed below. For example, the chassis **802** may be provided by a circuit board, although other chassis are envisioned as falling within the scope of the present disclosure as well. A plurality of connectors **802a**, **802b**, **802c**, and up to **802d** are included on the chassis **802**, and one of skill in the art in possession of the present disclosure will appreciate how the connector **802a** and the connectors **802b-802d** may be coupled to each other via the chassis **802**. For example, in embodiments in which the chassis **802** is a circuit board, that circuit board may include traces, wiring, couplings, and/or other circuit board subsystems that couple the connector **802a** to the connectors **802b-802d**. However, while a specific modular temperature measurement subsystem **800** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that modular intermediary subsystems (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the modular temperature measurement subsystem **800**) may include a variety of components and/or component configurations for providing the functionality discussed below while remaining within the scope of the present disclosure as well.

(34) Referring now to FIG. 9, an embodiment of a modular intermediary subsystem provided by a modular Peripheral Component Interconnect express (PCIe) connect subsystem **900** is illustrated that may be included in the modular test system **202** discussed above with reference to FIG. 2. In the illustrated embodiment, the modular PCIe connect subsystem **900** includes a chassis **902** that supports the components of the modular PCIe connect subsystem **900**, only some of which are illustrated and discussed below. For example, the chassis **902** may be provided by a circuit board, although other chassis are envisioned as falling within the scope of the present disclosure as well. A plurality of connectors **904a**, **904b**, and **904c** are included on the chassis **902**, and one of skill in the art in possession of the present disclosure will appreciate how the connectors **904a**, **904b**, and **904c** may be coupled to each other via the chassis **902**. For example, in embodiments in which the chassis **902** is a circuit board, that circuit board may include traces, wiring, couplings, and/or other circuit board subsystems that couple the connectors **904a**, **904b**, and **904c** to each other.

(35) In the specific examples described below, the chassis **902**/circuit board includes a first modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit communications between any of the connectors **904a**, **904b**, and **904c** during testing operations. Furthermore, in some embodiments, the

chassis **902**/circuit board may include a second modular intermediary board subsystem that may be configured to provide processing functionality that performs PCIe data routing functions on data transmitted between any of the connectors **904a**, **904b**, and/or **904c**.

(36) As such, while not explicitly illustrated in FIG. **9**, in some embodiments the chassis **902** may support a processing system (not illustrated, but which may include the processor **102** discussed above with reference to FIG. **1**) and a memory system (not illustrated, but which may include the memory **114** discussed above with reference to FIG. **1**) that is coupled to the processing system and that includes instructions that, when executed by the processing system, cause the processing system to provide a PCIe data routing engine that is configured to perform the functionality of the PCIe data routing engines and/or modular PCIe connect subsystems discussed below, and that one of skill in the art in possession of the present disclosure will appreciate may be coupled to any or all of the connectors **904a-904c**. However, while a specific modular PCIe connect subsystem **900** has been illustrated and described, one of skill in the art in possession of the present disclosure will recognize that modular intermediary subsystems (or other devices operating according to the teachings of the present disclosure in a manner similar to that described below for the modular PCIe connect subsystem **900**) may include a variety of components and/or component configurations for providing the functionality discussed below while remaining within the scope of the present disclosure as well.

(37) Referring now to FIG. **10**, an embodiment of a method **1000** for modular testing of a tested subsystem is illustrated. As discussed below, the systems and methods of the present disclosure provide a modular test system with modular intermediary subsystems that may be provided to couple a modular controller subsystem to a modular tested subsystem, with any of the modular intermediary subsystems configured to perform a testing function that is not available in the modular controller subsystem during testing operations. For example, the modular test system of the present disclosure may include a modular controller subsystem coupled to a modular tested subsystem by a modular intermediary subsystem. The modular intermediary subsystem transmits communications between the modular controller subsystem and the modular tested subsystem during testing operations performed by the modular controller subsystem, and performs at least one testing function that is not available in the modular controller subsystem during the testing operations performed by the modular controller subsystem. As such, the modular tested subsystem may have different usages and functions tested by interchanging different modular intermediary subsystems between the modular controller subsystem and that modular tested subsystem and using those modular intermediary subsystems to perform testing functions during testing operations, such as power measurement functions and data debug functions, that are not available in the modular controller subsystem that is performing those testing operations.

(38) In some embodiments the method **1000** may be preceded by first testing operations being performed by the modular controller subsystem **300** with the modular tested subsystem. As will be appreciated by one of skill in the art in possession of the present disclosure, modular tested subsystems may be provided with any of a variety of desired operating characteristics and/or conditions prior to any of the testing operations discussed below. For example, with reference to the modular media subsystem **400** having the media devices **404** that may be provided by storage media such as NAND storage subsystems, those NAND storage subsystems may be provided at some Program/Erase (P/E) cycle level by, for example, writing and erasing data to NAND blocks in the NAND storage subsystems until a desired P/E cycle level has been reached, may have data written to the NAND blocks in the NAND storage subsystems and then those NAND storage subsystems may be subjected to elevated temperatures (e.g., during a “bake” operation) in order to accelerate data retention effects, and/or may be subjected too other techniques that provide desired operating characteristics/conditions known in the art.

(39) With reference to FIG. **11**, one of the testing device(s) **204** discussed above with reference to FIG. **2** may be coupled to the connector **306a** on the modular controller subsystem **300** by, for

example, connecting a cable between the testing device **204** and the connector **306a** on the modular controller subsystem **300**, directly connecting the connector **306a** on the modular controller subsystem **300** to a corresponding connector on the testing device **204**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, the modular media subsystem **400** may be coupled to the modular controller subsystem **300** by connecting the connector **406** on the modular media subsystem **400** to the connector **306b** on the modular controller subsystem **300**. Subsequently, the testing device **204** and the modular controller device **300** may operate to perform first testing operations with the modular media subsystem **400**. As will be appreciated by one of skill in the art in possession of the present disclosure, different modular media subsystems with different configurations (e.g., different media packaging, different Ball Grid Array (BGA) styles, etc.), media device types (e.g., different NAND types), etc. may be provided with the modular controller subsystem **300** as illustrated in FIG. **11** and tested as described below while remaining within the scope of the present disclosure.

(40) To provide a specific example, the first testing operations performed by the testing device **204** and the modular controller device **300** with the modular media subsystem **400** may include high-speed data transmission testing operations that may include a user generating high-speed data transmission testing operation instructions using the testing device **204**, the testing device **204** providing those high-speed data transmission testing operation instructions to the controller device **304** on the modular controller subsystem **300**. The controller device **304** may then convert those high-speed data transmission testing operation instructions to high-speed data transmission testing operation commands and transmit those high-speed data transmission testing operation commands to any of the media device(s) **404** on the modular media subsystem **400**, or may generate data and transmit that data to any of the media device(s) **404** on the modular media subsystem **400**, or may receive data from any of the media device(s) **404** on the modular media subsystem **400**, and/or may perform any other high-speed data transmission operations that would be apparent to one of skill in the art in possession of the present disclosure. However, while high-speed data transmission operations are described and referenced in the examples below, one of skill in the art in possession of the present disclosure will appreciate how the modular controller subsystem **300** may perform a variety of different first testing operations when directly connected to the modular media subsystem **400**/modular tested subsystem as illustrated in FIG. **11** while remaining within the scope of the present disclosure as well.

(41) In some specific examples, the first testing operations performed by the modular controller subsystem **300** with the modular media subsystem **400**/modular tested subsystem may experience issues that can be analyzed by further testing that requires testing functionality that is not available in the modular controller subsystem **300**. For example, with reference to the high-speed data transmission testing operation discussed above, a command-fail issue may result in which data requested from a media device **404** is not received back by the controller device **304** on the modular controller subsystem **300** or the testing device **204**, and may require debug data analyzing functionality that allows particular communication/signal data exchanged with the media devices **404** on the modular media subsystem **400** to be identified and analyzed, and that is not available on the modular controller subsystem **300**. For example, as discussed above, the components and/or configurations required to perform the debug data analyzing functionality may, if provided on the modular controller subsystem **300**, prevent or hinder the performance of the high-speed data transmission testing functionality discussed above, and thus that functionality may not be provided on the modular controller subsystem **300**.

(42) Similarly, an issue may result with the high-speed data transmission testing operations, and may require power measurement functionality discussed above that allows power consumption by the media devices **404** on the modular media subsystem **400** to be identified and analyzed, and that is not available on the modular controller subsystem **300**. For example, as discussed above, the components and/or configurations required to perform the power measurement functionality may, if

provided on the modular controller subsystem **300**, prevent or hinder the performance of the high-speed data transmission testing functionality discussed above, and thus that functionality may not be provided on the modular controller subsystem **300**. However, while the need for functionality not available in the modular controller subsystem **300** has been described as being provided by an modular intermediary subsystem in response to issues that occur during testing operations using functionality provided by the modular controller subsystem **300**, the functionality that is not available in the modular controller subsystem **300** may be required absent an issue occurring during testing operations using functionality provided by the modular controller subsystem **300** while remaining within the scope of the present disclosure as well. Furthermore, while particular functionality that is not available in the modular controller subsystem **300** is described herein, one of skill in the art in possession of the present disclosure will appreciate how any functionality not available in the modular controller subsystem **300** may be needed for testing operations in some embodiments while remaining within the scope of the present disclosure.

(43) The method **1000** begins at block **1002** where a modular intermediary subsystem connects to a modular controller subsystem and a modular tested subsystem. As will be appreciated by one of skill in the art in possession of the present disclosure, any of a variety of modular intermediary subsystems may be utilized to connect the modular controller subsystem to the modular tested subsystem in order to, for example, perform some testing functionality that is not available in the modular controller subsystem and required based on issues resulting from testing operations that were previously performed by the modular controller subsystem with the modular tested subsystem, or perform some testing functionality that is not available in the modular controller subsystem regardless of any testing operations that were previously performed by the modular controller subsystem with the modular tested subsystem.

(44) For example, with reference to FIG. **12**, at block **1002** one of the testing device(s) **204a** discussed above with reference to FIG. **2** may be coupled to the connector **306a** on the modular controller subsystem **300** by, for example, connecting a cable between the testing device **204a** and the connector **306a** on the modular controller subsystem **300**, directly connecting the connector **306a** on the modular controller subsystem **300** to a corresponding connector on the testing device **204a**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, the modular media subsystem **400** may be coupled to the modular controller subsystem **300** by connecting the connector **406** on the modular media subsystem **400** to the connector **504b** on the modular power measuring subsystem **500**, and connecting the connector **306b** on the modular controller subsystem **300** to the connector **504a** on the modular power measuring subsystem **500**.

(45) Furthermore, as discussed below, in some embodiments another of the testing device(s) **204b** discussed above with reference to FIG. **2** may be coupled to the connector **504c** on the modular power measuring subsystem **500** by, for example, connecting a cable between the testing device **204b** and the connector **504c** on the modular power measuring subsystem **500**, directly connecting the connector **504c** on the modular power measuring subsystem **500** to a corresponding connector on the testing device **204b**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. However, one of skill in the art in possession of the present disclosure will appreciate how the connection of the testing device **204b** to the modular power measuring subsystem **500** is optional (e.g., as indicated by the dashed lines used for the testing device **204b**), as the functionality of the testing device **204b** may be integrated with the modular power measuring subsystem **500**.

(46) For example, the modular power measuring subsystem **500** may be utilized to couple the modular media subsystem **400** to the modular controller subsystem **300** in order to perform power measurement functionality that is not available in the modular controller subsystem **300** and that may include measuring power consumed by the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300**. However, in embodiments in

which the modular power measuring subsystem **500** is not configured to perform the power measurement functionality discussed above (e.g., when that power measurement functionality requires components/configurations that may not be supported on the chassis **502** of the modular power measuring subsystem **500**), the modular power measuring subsystem **500** may be utilized to couple the modular media subsystem **400** to the modular controller subsystem **300** in order to perform power measurement data routing functionality that is not available in the modular controller subsystem **300** and that may include routing power consumption data generated by the modular media subsystem **400** to the testing device **204b**, with that testing device **204b** then using that power consumption data to measure power consumed by the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300**.

(47) Furthermore, while some examples herein describe the performance of the power measurement functionality or the power measurement data routing functionality by the modular power measuring subsystem **500** in response to issue with previous testing operations performed by the modular controller subsystem **300** with the modular media subsystem **400**, the power measurement functionality or the power measurement data routing functionality may be performed by the modular power measuring subsystem **500** regardless of any other testing operations being performed on the modular media subsystem **400** while remaining within the scope of the present disclosure as well.

(48) In another example, with reference to FIG. **13**, at block **1002** one of the testing device(s) **204a** discussed above with reference to FIG. **2** may be coupled to the connector **306a** on the modular controller subsystem **300** by, for example, connecting a cable between the testing device **204a** and the connector **306a** on the modular controller subsystem **300**, directly connecting the connector **306a** on the modular controller subsystem **300** to a corresponding connector on the testing device **204a**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, the modular media subsystem **400** may be coupled to the modular controller subsystem **300** by connecting the connector **406** on the modular media subsystem **400** to the connector **604b** on the modular debug data analyzing subsystem **600**, and connecting the connector **306b** on the modular controller subsystem **300** to the connector **604a** on the modular debug data analyzing subsystem **600**.

(49) Furthermore, as discussed below, in some embodiments another of the testing device(s) **204b** discussed above with reference to FIG. **2** may be coupled to the connector **604c** on the modular debug data analyzing subsystem **600** by, for example, connecting a cable between the testing device **204b** and the connector **604c** on the modular debug data analyzing subsystem **600**, directly connecting the connector **604c** on the modular debug data analyzing subsystem **600** to a corresponding connector on the testing device **204b**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. However, one of skill in the art in possession of the present disclosure will appreciate how the connection of the testing device **204b** to the modular debug data analyzing subsystem **600** is optional (e.g., as indicated by the dashed lines used for the testing device **204b**), as the functionality of the testing device **204b** may be integrated with the modular debug data analyzing subsystem **600**.

(50) For example, the modular debug data analyzing subsystem **600** may be utilized to couple the modular media subsystem **400** to the modular controller subsystem **300** in order to perform data debug analyzing functionality that is not available in the modular controller subsystem **300** and that may include analyzing debug data generated by the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300**. However, in embodiments in which the modular debug data analyzing subsystem **600** is not configured to perform the debug data analyzing functionality discussed above (e.g., when that debug data analyzing functionality requires components/configurations that may not be supported on the chassis **602** of the modular debug data analyzing subsystem **600**), the modular debug data analyzing subsystem **600** may be utilized to couple the modular media subsystem **400** to the modular controller subsystem **300** in

order to perform debug data routing functionality that is not available in the modular controller subsystem **300** and that may include routing debug data generated by the modular media subsystem **400** to the testing device **204b**, with that testing device **204b** then analyzing debug data generated by the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300**.

(51) Furthermore, while some examples herein describe the performance of the debug data analyzing functionality or the debug data routing functionality by the modular debug data analyzing subsystem **600** in response to issue with previous testing operations performed by the modular controller subsystem **300** with the modular media subsystem **400**, the debug data analyzing functionality or the debug data routing functionality may be performed by the modular debug data analyzing subsystem **600** regardless of any other testing operations being performed on the modular media subsystem **400** while remaining within the scope of the present disclosure as well.

(52) In another example, with reference to FIG. **14**, at block **1002** one of the testing device(s) **204** discussed above with reference to FIG. **2** may be coupled to the connector **306a** on the modular controller subsystem **300** by, for example, connecting a cable between the testing device **204** and the connector **306a** on the modular controller subsystem **300**, directly connecting the connector **306a** on the modular controller subsystem **300** to a corresponding connector on the testing device **204**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, a plurality of modular media subsystems **400** may be coupled to the modular controller subsystem **300** by connecting the connector **406** on each respective modular media subsystem **400** to one of the connectors **704b** and **704c** on the modular expansion subsystem **700**, and connecting the connector **306b** on the modular controller subsystem **300** to the connector **704a** on the modular expansion subsystem **700**. As will be appreciated by one of skill in the art in possession of the present disclosure, the modular expansion subsystem **700** may enable testing operations in which a maximum quantity of modular media subsystems or media devices are tested, particularly when expansion-based performance losses are acceptable in those testing operations (e.g., long-term data retention testing operations where scanning requirements are relatively low).

(53) In some embodiments, the modular expansion subsystem **700** performs an expansion functionality that is not available in the modular controller subsystem **300** and that allows the plurality of modular media subsystems **400** to be coupled to the modular controller subsystem **300** in order to allow the modular controller subsystem **300** to perform testing operations on each of the plurality of modular media subsystems **400**. Furthermore, while not illustrated or described herein in detail, one of skill in the art in possession of the present disclosure will appreciate how the other modular intermediary subsystems described herein (e.g., the modular power measurement subsystem **500** and the modular data debug analyzer subsystem **600** discussed above, the modular PCIe connect subsystem **900** described below) may be utilized to couple any of the modular media subsystems **400** to the modular expansion subsystem **700** in a manner similar to that described above with regard to their coupling the modular media subsystem **400** to the modular controller subsystem **300**.

(54) In another example, with reference to FIG. **15**, at block **1002** one of the testing device(s) **204a** discussed above with reference to FIG. **2** may be coupled to the connector **306a** on the modular controller subsystem **300** by, for example, connecting a cable between the testing device **204a** and the connector **306a** on the modular controller subsystem **300**, directly connecting the connector **306a** on the modular controller subsystem **300** to a corresponding connector on the testing device **204a**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, the modular media subsystem **400** may be coupled to the modular controller subsystem **300** by connecting the connector **406** on the modular media subsystem **400** to the connector **904b** on the modular PCIe connect subsystem **900**, and

connecting the connector **306b** on the modular controller subsystem **300** to the connector **904a** on the modular PCIe connect subsystem **900**.

(55) Furthermore, as discussed below, controller subsystem **1500** may be coupled to the connector **904c** on the modular PCIe connect subsystem **900** by, for example, connecting a cable between the controller subsystem **1500** and the connector **904c** on the modular PCIe connect subsystem **900**, directly connecting the connector **904c** on the modular PCIe connect subsystem **900** to a corresponding connector on the controller subsystem **1500**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. In some embodiments, the modular PCIe connect subsystem **900** may be utilized to couple the modular media subsystem **400**, the modular controller subsystem **300**, and the controller subsystem **1500** to each other in order to allow the modular controller subsystem **300** and the controller subsystem **1500** to perform testing operations on the modular media subsystem **400**. For example, the controller subsystem **1500** is described in specific examples below as a developmental controller that is configured with limiting testing functionality relative to the modular controller subsystem **300**, and the controller subsystem **1500** utilizes the modular controller subsystem **300** to perform at least some testing operations with the modular media subsystem **400**. However, while the connection of a particular controller subsystem to the modular PCIe connect subsystem **900** is described, one of skill in the art in possession of the present disclosure will appreciate how the connection of other PCIe devices to the modular PCIe connect subsystem **900** will fall within the scope of the present disclosure as well.

(56) As such, one of skill in the art in possession of the present disclosure will recognize that a user of the modular test system of the present disclosure may have a plurality of the modular intermediary subsystems that are each configured to perform a respective testing function that is not available from the modular controller subsystem, and may utilize any of those modular intermediary subsystems to couple the modular controller subsystem to the modular tested subsystem depending on the type of testing operations that are desired. However, while a few specific modular intermediary subsystems are illustrated and described herein, one of skill in the art in possession of the present disclosure will appreciate how other modular intermediary subsystems that provide other testing functionality not available in the modular controller subsystem will fall within the scope of the present disclosure as well. For example, a modular error testing subsystem may be provided as a modular intermediary subsystem and may be configured to inject errors into NAND storage subsystems in order to, for example, test error correction capabilities of the NAND storage subsystems, add delays in some signals so that some data is received later than other data in order to test the ability of the NAND storage subsystems to manage such delays (i.e., based on an allowable “skew” in those signals), and/or perform other error-based testing that would be apparent to one of skill in the art in possession of the present disclosure.

(57) The method **1000** then proceeds to block **1004** where the modular intermediary subsystem transmits communications between the modular controller subsystem and the modular tested subsystem during testing operations performed by the modular controller subsystem. In an embodiment, at block **1004**, a testing device and the modular controller subsystem may perform testing operations that include exchanging communications with the modular tested subsystem, and the modular intermediary subsystem that couples the modular controller subsystem to the modular tested subsystem will transmit those communications between the modular controller subsystem and the modular tested subsystem. For example, a user may generate testing operation instructions using the testing device **204**, the testing device **204** may provide those testing operation instructions to the controller device **304** on the modular controller subsystem **300**, and the controller device **304** may convert those testing operation instructions to testing operation commands and transmit those testing operation commands to the modular intermediary subsystem such that the modular intermediary subsystem transmits those testing operation commands to any of the media device(s) **404** on the modular media subsystem **400**.

(58) Similarly, a user may generate testing operation instructions using the testing device **204**, the testing device **204** may provide those testing operation instructions to the controller device **304** on the modular controller subsystem **300**, and the controller device **304** may generate data in response to those testing operation instructions and transmit that data to the modular intermediary subsystem such that the modular intermediary subsystem transmits that data to any of the media device(s) **404** on the modular media subsystem **400**. Furthermore, any of the media devices **404** on the modular media subsystem **400** may generate data in response to data or testing operating commands received via the modular intermediary subsystem from the modular controller device **300**, and transmit that data to the modular intermediary subsystem such that the modular intermediary subsystem transmits that data to the controller device **304** on the modular controller subsystem **300**.

(59) For example, with reference to the modular power measuring subsystem **500** discussed above with reference to FIGS. **5** and **12**, the chassis **502**/circuit board in the modular power measuring subsystem **500** may include a first modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit communications between the connectors **504a** and **504b** during testing operations, and thus at block **1004** the first modular intermediary board subsystem may be utilized to transmit the communications exchanged between the modular controller subsystem **300** and the modular media subsystem **400** during the testing operations performed by the testing device **204a** and the modular controller subsystem **300**. For example, the circuitry provided by the first modular intermediary board subsystem on the modular power measuring subsystem **500** may provide “through” connections that are configured to transmit power and data to enable “normal” operation of the modular media subsystem **400**.

(60) In another example, with reference to the modular debug data analyzing subsystem **600** discussed above with reference to FIGS. **6** and **13**, the chassis **602**/circuit board in the modular debug data analyzing subsystem **600** may include a first modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit communications between the connectors **604a** and **604b** during testing operations, and thus at block **1004** the first modular intermediary board subsystem may be utilized to transmit the communications exchanged between the modular controller subsystem **300** and the modular media subsystem **400** during the testing operations performed by the testing device **204a** and the modular controller subsystem **300**. For example, the circuitry provided by the first modular intermediary board subsystem on the modular debug data analyzing subsystem **600** may provide “through” connections that are configured to transmit power and data to enable “normal” operation of the modular media subsystem **400**.

(61) In another example, with reference to the modular expansion subsystem **700** discussed above with reference to FIGS. **7** and **14**, the chassis **702**/circuit board in the modular expansion subsystem **700** may include a first modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit communications between the connector **704a** and the connector **704b** during testing operations, and thus at block **1004** the first modular intermediary board subsystem may be utilized to transmit the communications exchanged between the modular controller subsystem **300** and the modular media subsystem **400** that is connected to the connector **704b** during the testing operations performed by the testing device **204** and the modular controller subsystem **300**. For example, the circuitry provided by the first modular intermediary board subsystem on the modular expansion subsystem **700** may provide “through” connections that are configured to transmit power and data to enable “normal” operation of the modular media subsystems **400**.

(62) In another example, with reference to the modular expansion subsystem **900** discussed above with reference to FIGS. **9** and **15**, the chassis **902**/circuit board in the modular PCIe connect subsystem **900** may include a first modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit

communications between the connectors **904a** and **904b** during testing operations, and thus at block **1004** the first modular intermediary board subsystem may be utilized to transmit the communications exchanged between the modular controller subsystem **300** and the modular media subsystem **400** during the testing operations performed by the testing device **204** and the modular controller subsystem **300**. For example, the circuitry provided by the first modular intermediary board subsystem on the modular expansion subsystem **900** may provide “through” connections that are configured to transmit power and data to enable “normal” operation of the modular media subsystems **400**.

(63) The method **1000** then proceeds to block **1006** where the modular intermediary subsystem performs testing function(s) that are not available in the modular controller subsystem during the testing operations performed by the modular controller subsystem. In an embodiment, at block **1006** and while the testing device and the modular controller subsystem are performing the testing operations discussed above, the modular intermediary subsystem that couples the modular controller subsystem to the modular tested subsystem may perform at least one testing operation that is not available in the modular controller subsystem.

(64) For example, with reference to the modular power measuring subsystem **500** discussed above with reference to FIGS. **5** and **12**, the chassis **502**/circuit board in the modular power measuring subsystem **500** may include a second modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit data received at the connector **504b** to the connector **504c**, and at block **1006** that second modular intermediary board subsystem may perform power measurement data routing function to route power measurement data generated by the media devices **404** in the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300** to the testing device **204b**, with the testing device **204b** performing a power management function to, for example, measure power consumed by the media devices **404** in the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300**. However, in other embodiments, the second modular intermediary board subsystem on the chassis **502**/circuit board may provide a power measuring engine that is configured to provide processing functionality that performs the power management function to, for example, measure power consumed by the media devices **404** in the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300**, and may generate corresponding power consumption data and transmit that power consumption data via the connector **504a** to the controller device **304** (which may then provide that power consumption data to the testing device **204a**).

(65) In another example, with reference to the modular debug data analyzing subsystem **600** discussed above with reference to FIGS. **6** and **13**, the chassis **602**/circuit board in the modular debug data analyzing subsystem **600** may include a second modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit data received at the connector **604b** to the connector **604c**, and at block **1006** that second modular intermediary board subsystem may perform debug data routing function to route debug data generated by the media devices **404** in the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300** to the testing device **204b**, with the testing device **204b** performing a debug data analyzing function to, for example, analyze debug data generated by the media devices **404** in the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300**. However, in other embodiments, the second modular intermediary board subsystem on the chassis **602**/circuit board may provide a debug data analyzing engine that is configured to provide processing functionality that performs the debug data analyzing function to, for example, analyze debug data generated by the media devices **404** in the modular media subsystem **400** during the testing operations performed by the modular controller subsystem **300**, and may generate corresponding debug analysis data and transmit that debug analysis data via the connector **604a** to the controller

device **304** (which may then provide that debug analysis data to the testing device **204a**). As will be appreciated by one of skill in the art in possession of the present disclosure, the debug data analyzing functions discussed above may include analyzing timing operations, analyzing bus operations, determining whether data is being read/written at the right time, and/or other debug data analyzing functions that would be apparent to one of skill in the art in possession of the present disclosure.

(66) In another example, with reference to the modular expansion subsystem **700** discussed above with reference to FIGS. **7** and **14**, the chassis **702**/circuit board in the modular debug data analyzing subsystem **700** may include a second modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to route communications between the connector **704a** and the connector **704c**, and at block **1006** that second modular intermediary board subsystem may perform expansion communication routing function to transmit the communications exchanged between the modular controller subsystem **300** and media devices **404** in the modular media subsystem **400** connected to the connector **704c** during the testing operations performed by the modular controller subsystem **300**. Furthermore, in some embodiments, the second modular intermediary board subsystem on the chassis **702**/circuit board may provide an expansion communication routing engine that is configured to provide processing functionality that performs the expansion communication routing function to, for example, process and transmit the communications exchanged between the modular controller subsystem **300** and media devices **404** in the modular media subsystems **400** connected to the connectors **704b** and **704c** during the testing operations performed by the modular controller subsystem **300**.

(67) In another example, with reference to the modular PCIe connect subsystem **900** discussed above with reference to FIGS. **9** and **15**, the chassis **902**/circuit board in the modular PCIe connect subsystem **900** may include a second modular intermediary board subsystem that may be provided by circuitry (e.g., the traces, wiring, couplings, etc. discussed above) that is configured to transmit data between the connectors **904a**, **904b**, and **904c**, and at block **1006** that second modular intermediary board subsystem may perform PCIe connect communication routing functions to receive communications from a subsystem connected to any of those connectors **904a-904c** during the testing operations performed by the modular controller subsystem **300** and route those communications to a subsystem connected to any other of those connectors. Furthermore, in some embodiments, the second modular intermediary board subsystem on the chassis **902**/circuit board may provide a PCIe connect communication routing engine that is configured to provide processing functionality that performs the PCIe connect communication routing functions to receive communications from a subsystem connected to any of those connectors **904a-904c** during the testing operations performed by the modular controller subsystem **300** and route those communications to a subsystem connected to any other of those connectors.

(68) To provide a specific example of the operation of the embodiment of the modular test system illustrated in FIG. **15**, the controller subsystem **1500** may be provided by a developmental controller that may not be fully programmed to perform all testing operations on the modular media subsystem **400**, while the modular controller subsystem **300** may be configured to perform all testing operations on the modular media subsystem **400**. As such, at block **1006**, the second modular intermediary board subsystem in the modular PCIe connect subsystem **900** may route some communications between the controller subsystem **1500** and the media devices **404** in the modular media subsystem **400** (i.e., communications associated with the testing operations the controller subsystem **1500** is configured to perform), while routing some communications (e.g., relatively high-level commands) between the controller subsystem **1500** and the controller device **304** in the modular controller subsystem **300** (i.e., communications associated with the testing operations the controller subsystem **1500** is not configured to perform).

(69) As will be appreciated by one of skill in the art in possession of the present disclosure, when

the modular controller subsystem **300** receives communication from the controller subsystem **1500** associated with testing operations that the controller subsystem **1500** is not configured to perform, the modular controller subsystem **300** may transmit corresponding communications (e.g., relatively low-level commands such as Firmware Translation Layer (FTL) commands, NAND commands, etc.) to the media devices **404** in the modular media subsystem **400** that are configured to perform the testing operations that the controller subsystem **1500** is not configured to perform. Thus, some embodiments of the modular PCI connect subsystem **900** allow a developmental controller to be tested with media devices, while using the modular controller subsystem **300** to provide for any testing operations the developmental controller is not configured to perform. Furthermore, one of skill in the art in possession of the present disclosure will appreciate that the operations discussed above (e.g., the NAND/FTL abstraction operations for the controller subsystem **1500** using the modular controller subsystem **300**) may allow for the testing of developmental controller functionality by leveraging “legacy” low-level NAND firmware on an existing “legacy” modular controller subsystem.

(70) As will be appreciated by one of skill in the art in possession of the present disclosure, in embodiments in which one of the modular intermediary subsystems of the present disclosure was utilized subsequent to testing operations performed by the modular controller subsystem **300** in which an issue was experienced that required testing functions that are enabled by that modular intermediary subsystem and that are not available in the modular controller subsystem **300**, that modular intermediary subsystem may be removed from the modular test system **200** so that the modular controller subsystem **300** and modular media subsystem **400** may again be directly connected in order to complete the testing operations (e.g., the high-speed data transmission testing operations discussed above) being performed by the modular controller subsystem **300**.

(71) With reference to FIG. **16**, an embodiment of the use of the modular temperature measurement subsystem **800** discussed above with reference to FIG. **8** is illustrated. As can be seen, one of the testing device(s) **204** discussed above with reference to FIG. **2** may be coupled to the connector **802a** on the modular temperature measurement subsystem **800** by, for example, connecting a cable between the testing device **204** and the connector **802a** on the modular temperature measurement subsystem **800**, directly connecting the connector **802a** on the modular temperature measurement subsystem **800** to a corresponding connector on the testing device **204**, and/or providing other couplings that would be apparent to one of skill in the art in possession of the present disclosure. Furthermore, a plurality of the modular media subsystems **400** may be coupled to the modular temperature measurement subsystem **800** by connecting the connector **406** on each respective modular media subsystem **400** to one of each of the connectors **802b** and up to **802d** on the modular temperature measurement subsystem **800**.

(72) In an embodiment, the modular temperature measurement subsystem **800** may be configured to receive temperature measurements (e.g., provided by temperature sensors included with the media devices **404** on the modular media subsystems **400**) and provide those temperature measurements to the testing device **204**. As will be appreciated by one of skill in the art in possession of the present disclosure, the temperature measurement functionality described above for the modular temperature measurement subsystem **800** may be performed as part of testing operations that occur without the need for the modular controller subsystem **300** or its functionality discussed above. For example, the modular temperature measurement subsystem **800** may not include a conventional controller or controller functionality, and instead may be provided by a “lightweight” circuit board that is configured to simply read the temperature sensors included with the media devices **404** on the modular media subsystems **400**, and provide the temperature data obtained to the testing device **204**. As such, the modular temperature measurement subsystem **800** may be provided in a temperature-controlled environment (e.g., a test chamber/“oven”) and used to perform temperature testing on the modular media subsystem **400**.

(73) Thus, systems and methods have been described that provide a modular test system with

modular intermediary subsystems that may be provided to couple a modular controller subsystem to a modular tested subsystem, with any of the modular intermediary subsystems configured to perform a testing function that is not available in the modular controller subsystem during testing operations. For example, the modular test system of the present disclosure may include a modular controller subsystem coupled to a modular tested subsystem by a modular intermediary subsystem. The modular intermediary subsystem transmits communications between the modular controller subsystem and the modular tested subsystem during testing operations performed by the modular controller subsystem, and performs at least one testing function that is not available in the modular controller subsystem during the testing operations performed by the modular controller subsystem. As such, the modular tested subsystem may have different usages and functions tested by interchanging different modular intermediary subsystems between the modular controller subsystem and that modular tested subsystem, and using those modular intermediary subsystems to perform testing functions that are not available in the modular controller subsystem that is performing those testing operations. In some embodiments, the modular intermediary subsystems of the present disclosure may be interchangeable in modular test systems in order to test NAND storage subsystems at “normal” maximum line rate conditions, provide full debug capability as needed, provide power measurement, and provide offline NAND storage subsystem temperature measurement, thus decoupling upgrade developmental pathways for controllers and NAND storage subsystems, and enabling both small and large scale NAND storage subsystem testing.

(74) Although illustrative embodiments have been shown and described, a wide range of modification, change and substitution is contemplated in the foregoing disclosure and in some instances, some features of the embodiments may be employed without a corresponding use of other features. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the embodiments disclosed herein.

Claims

1. A modular test system, comprising: a modular tested subsystem that includes a plurality of NAND storage subsystems; a modular controller subsystem that is configured to connect directly to the modular tested subsystem and, while connected directly to the modular tested subsystem, to perform data transmission testing operations on one or more of the plurality of NAND storage subsystems included on the modular tested subsystem; a power measuring modular intermediary subsystem that is configured to connect the modular controller subsystem indirectly to the modular tested subsystem and, while connecting the modular controller subsystem indirectly to the modular tested subsystem, to: transmit power measurement testing communications between the modular controller subsystem and the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to cause the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to perform power measurement testing operations; and perform at least one power measurement testing function that is not available in the modular controller subsystem and that provides for measurement of power consumed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations; and a debug data analyzing modular intermediary subsystem that is configured to connect the modular controller subsystem indirectly to the modular tested subsystem and, while connecting the modular controller subsystem indirectly to the modular tested subsystem, to: transmit data debug analyzing communications between the modular controller subsystem and the modular tested subsystem to cause the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to perform debug data analyzing testing operations; and perform at least one debug data analyzing testing function that is not available in the modular controller subsystem and that provides for analysis of debug data generated by the one or more of the plurality of NAND storage subsystems included on the

modular tested subsystem during the debug data analyzing testing operations.

2. The modular test system of claim 1, wherein the at least one power measurement testing function performed by the power measuring modular intermediary subsystem includes a power measurement function to measure the power consumed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations.

3. The modular test system of claim 1, wherein the at least one power measurement testing function performed by the power measuring modular intermediary subsystem includes a power measurement data routing function to route power consumption data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations to a testing device that is coupled to the power measuring modular intermediary subsystem.

4. The modular test system of claim 1, wherein the at least one debug data analyzing testing function performed by the debug data analyzing modular intermediary subsystem includes a debug data analyzing function to analyze the debug data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations.

5. The modular test system of claim 1, wherein the at least one debug data analyzing testing function performed by the debug data analyzing modular intermediary subsystem includes a debug data routing function to route the debug data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to a testing device that is coupled to the debug data analyzing modular intermediary subsystem.

6. The modular test system of claim 1, wherein the modular tested subsystem is a first modular tested subsystem that includes a plurality of first NAND storage subsystems, and wherein the modular test system further comprises: an expansion modular intermediary subsystem that is configured to connect the modular controller subsystem indirectly to the first modular tested subsystem and to a second modular tested subsystem that includes a plurality of second NAND storage subsystems and, while connecting the modular controller subsystem indirectly to the first modular tested subsystem and the second modular tested subsystem, to: transmit data transmission testing communications between the modular controller subsystem and one or more of the plurality of first NAND storage subsystems included on the first modular tested subsystem to cause the one or more of the plurality of first NAND storage subsystems included on the first modular tested subsystem to perform the data transmission testing operations; and transmit data transmission testing communications between the modular controller subsystem and one or more of the plurality of second NAND storage subsystems included on the second modular tested subsystem to cause the one or more of the plurality of second NAND storage subsystems included on the second modular tested subsystem to perform the data transmission testing operations.

7. A modular intermediary subsystem, comprising: a power measuring modular intermediary board including: a first power measuring modular intermediary connector that is configured to connect to a modular controller subsystem that is configured to be connected directly to a modular tested subsystem and, while connected directly to the modular tested subsystem, to perform data transmission testing operations on one or more of a plurality of NAND storage subsystems

included on the modular tested subsystem; a second power measuring modular intermediary connector that is configured to connect to the modular tested subsystem to indirectly connect the modular controller subsystem to the modular tested subsystem; a first power measuring modular intermediary board subsystem that is configured to: transmit power measurement testing communications between the modular controller subsystem and the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to cause the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to perform power measurement testing operations; and a second power measuring modular intermediary board subsystem that is configured to: perform at least one power measurement testing function that is not available in the modular controller subsystem and that provides for measurement of power consumed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations; and a debug data analyzing modular intermediary board including: a first debug data analyzing modular intermediary connector that is configured to connect to the modular controller subsystem; a second debug data analyzing modular intermediary connector that is configured to connect to the modular tested subsystem to indirectly connect the modular controller subsystem to the modular tested subsystem; a first debug data analyzing modular intermediary board subsystem that is configured to: transmit debug data analyzing testing communications between the modular controller subsystem and the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to cause the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to perform debug data analyzing testing operations; and a second debug data analyzing modular intermediary board subsystem that is configured to: perform at least one debug data analyzing testing function that is not available in the modular controller subsystem and that provides for analysis of debug data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations.

8. The modular intermediary subsystem of claim 7, wherein the at least one power measurement testing function performed by the second power measuring modular intermediary board subsystem includes a power measurement function to measure the power consumed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations.

9. The modular intermediary subsystem of claim 7, further comprising: a third power measuring modular intermediary connector that is included on the power measuring modular intermediary board, wherein the at least one power measurement testing function performed by the second power measuring modular intermediary board subsystem includes a power measurement data routing function to route power consumption data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations to a testing device that is connected to the third power measuring modular intermediary connector.

10. The modular intermediary subsystem of claim 7, wherein the at least one debug data analyzing testing function performed by the second debug data analyzing modular intermediary board subsystem includes a debug data analyzing function to analyze the debug data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations.

11. The modular intermediary subsystem of claim 7, further comprising: a third debug data analyzing modular intermediary connector that is included on the debug data analyzing modular intermediary board, wherein the at least one debug data analyzing testing function includes a debug data routing function to route the debug data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to a testing device that is connected to the third debug data analyzing modular intermediary connector.

12. The modular intermediary subsystem of claim 7, wherein the modular tested subsystem is a first modular tested subsystem including a plurality of first NAND storage subsystems, and wherein the modular intermediary subsystem further comprises: an expansion modular intermediary board including: a first expansion modular intermediary connector that is configured to connect to the modular controller subsystem; a second expansion modular intermediary connector that is configured to connect to the first modular tested subsystem to indirectly connect the modular controller subsystem to the first modular tested subsystem; a third expansion modular intermediary connector that is configured to connect to a second modular tested subsystem including a plurality of second NAND storage subsystems to indirectly connect the modular controller subsystem to the second modular tested subsystem; and a data transmission modular intermediary board subsystem that is configured to: transmit data transmission testing communications between the modular controller subsystem and one or more of the plurality of first NAND storage subsystems included on the first modular tested subsystem to cause the one or more of the plurality of first NAND storage subsystems included on the first modular tested subsystem to perform the data transmission testing operations; and transmit data transmission testing communications between the modular controller subsystem and one or more of the plurality of second NAND storage subsystems included on the second modular tested subsystem to cause the one or more of the plurality of second NAND storage subsystems included on the second modular tested subsystem to perform the data transmission testing operations.

13. The modular intermediary subsystem of claim 7, wherein the modular controller subsystem is a first modular controller subsystem, and wherein the modular intermediary subsystem further comprises: a multi-controller modular intermediary board including: a first multi-controller modular intermediary connector that is configured to connect to the first modular controller subsystem; a second multi-controller modular intermediary connector that is configured to connect to a second modular controller subsystem; a third multi-controller modular intermediary connector that is configured to connect to the modular tested subsystem to indirectly connect each of the first modular controller subsystem and the second modular controller subsystem to the modular tested subsystem; and a multi-controller modular intermediary board subsystem that is configured to: transmit data transmission testing communications between the first modular controller subsystem and one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to cause the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to perform the data transmission testing operations; and forward commands received from the second modular controller subsystem to the first modular controller subsystem.

14. A method for modular testing of a tested subsystem, comprising: connecting, by a power measuring modular intermediary subsystem, to a modular controller subsystem that is configured to be connected directly to a modular tested subsystem and, while connected directly to the modular tested subsystem, to perform data transmission testing operations on one or more of a plurality of NAND storage subsystems included on the modular tested subsystem; connecting, by the power measuring modular intermediary subsystem, to the modular tested subsystem to indirectly connect the modular controller subsystem to the modular tested subsystem; transmitting, by the power measuring modular intermediary subsystem, power measurement testing communications between

the modular controller subsystem and the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to cause the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to perform power measurement testing operations; performing, by the power measuring modular intermediary subsystem, at least one power measurement testing function that is not available in the modular controller subsystem and that provides for measurement of power consumed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations; connecting, by a debug data analyzing modular intermediary subsystem, to the modular controller subsystem; connecting, by the debug data analyzing modular intermediary subsystem, to the modular tested subsystem to indirectly connect the modular controller subsystem to the modular tested subsystem; and transmitting, by the debug data analyzing modular intermediary subsystem, debug data analyzing testing communications between the modular controller subsystem and the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to cause the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to perform debug data analyzing testing operations; performing, by the debug data analyzing modular intermediary subsystem, at least one debug data analyzing testing function that is not available in the modular controller subsystem and that provides for analysis of debug data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations.

15. The method of claim 14, wherein performing the at least one power measurement testing function includes: performing a power measurement function to measure the power consumed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations.

16. The method of claim 14, wherein performing the at least one power measurement testing function includes: performing a power measurement data routing function to route power consumption data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the power measurement testing operations to a testing device that is connected to the power measuring modular intermediary subsystem.

17. The method of claim 14, wherein performing the at least one debug data analyzing testing function includes: performing a debug data analyzing function to analyze the debug data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations.

18. The method of claim 14, wherein performing the at least one debug data analyzing testing function includes: performing a debug data routing function to route the debug data generated by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem during the debug data analyzing testing operations performed by the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to a testing device that is connected to the debug data analyzing modular intermediary subsystem.

19. The method of claim 14, wherein the modular tested subsystem is a first modular tested subsystem including a plurality of first NAND storage subsystems, and wherein the method further comprises: connecting, by an expansion modular intermediary subsystem, to the modular controller subsystem; connecting, by the expansion modular intermediary subsystem, to the first modular tested subsystem to indirectly connect the modular controller subsystem to the first modular tested

subsystem; connecting, by the expansion modular intermediary subsystem, to a second modular tested subsystem that includes a plurality of second NAND storage subsystems to indirectly connect the modular controller subsystem to the second modular tested subsystem; transmitting, by the expansion modular intermediary subsystem, data transmission testing communications between the modular controller subsystem and one or more of the plurality of first NAND storage subsystems included on the first modular tested subsystem to cause the one or more of the plurality of first NAND storage subsystems included on the first modular tested subsystem to perform the data transmission testing operations; and transmitting, by the expansion modular intermediary subsystem, data transmission testing communications between the modular controller subsystem and one or more of the plurality of second NAND storage subsystems included on the second modular tested subsystem to cause the one or more of the plurality of second NAND storage subsystems included on the second modular tested subsystem to perform the data transmission testing operations.

20. The method of claim 14, wherein the modular controller subsystem is a first modular controller subsystem and the method further comprises: connecting, by a multi-controller modular intermediary subsystem, to the first modular controller subsystem; connecting, by the multi-controller modular intermediary subsystem, to a second modular controller subsystem; connecting, by the multi-controller modular intermediary subsystem, to the modular tested subsystem to indirectly connect each of the first modular controller subsystem and the second modular controller subsystem to the modular tested subsystem; transmitting, by the multi-controller modular intermediary subsystem, data transmission testing communications between the first modular controller subsystem and the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to cause the one or more of the plurality of NAND storage subsystems included on the modular tested subsystem to perform the data transmission testing operations; and forwarding, by the multi-controller modular intermediary subsystem, commands received from the second modular controller subsystem to the first modular controller subsystem.
